INCH-POUND

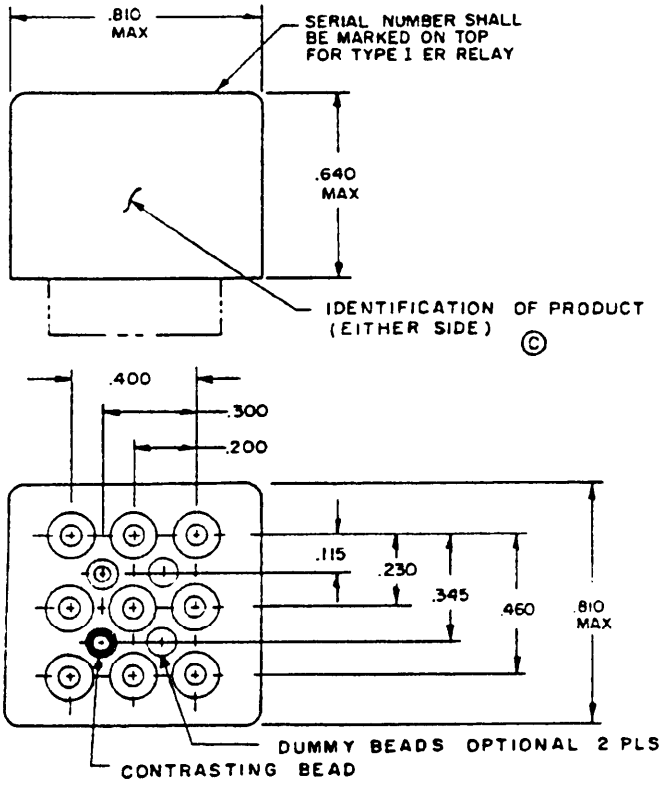
MIL-R-6106/29C
26 December 1989
SUPERSEDING
MIL-R-6106/29B
24 May 1985

MILITARY SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, (INCLUDING ESTABLISHED RELIABILITY (ER) TYPES), PERMANENT MAGNET DRIVE, LOW LEVEL TO 10 AMPERES, 3 PDT, ALL WELDED, HERMETICALLY SEALED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: (MIL-R-6106).



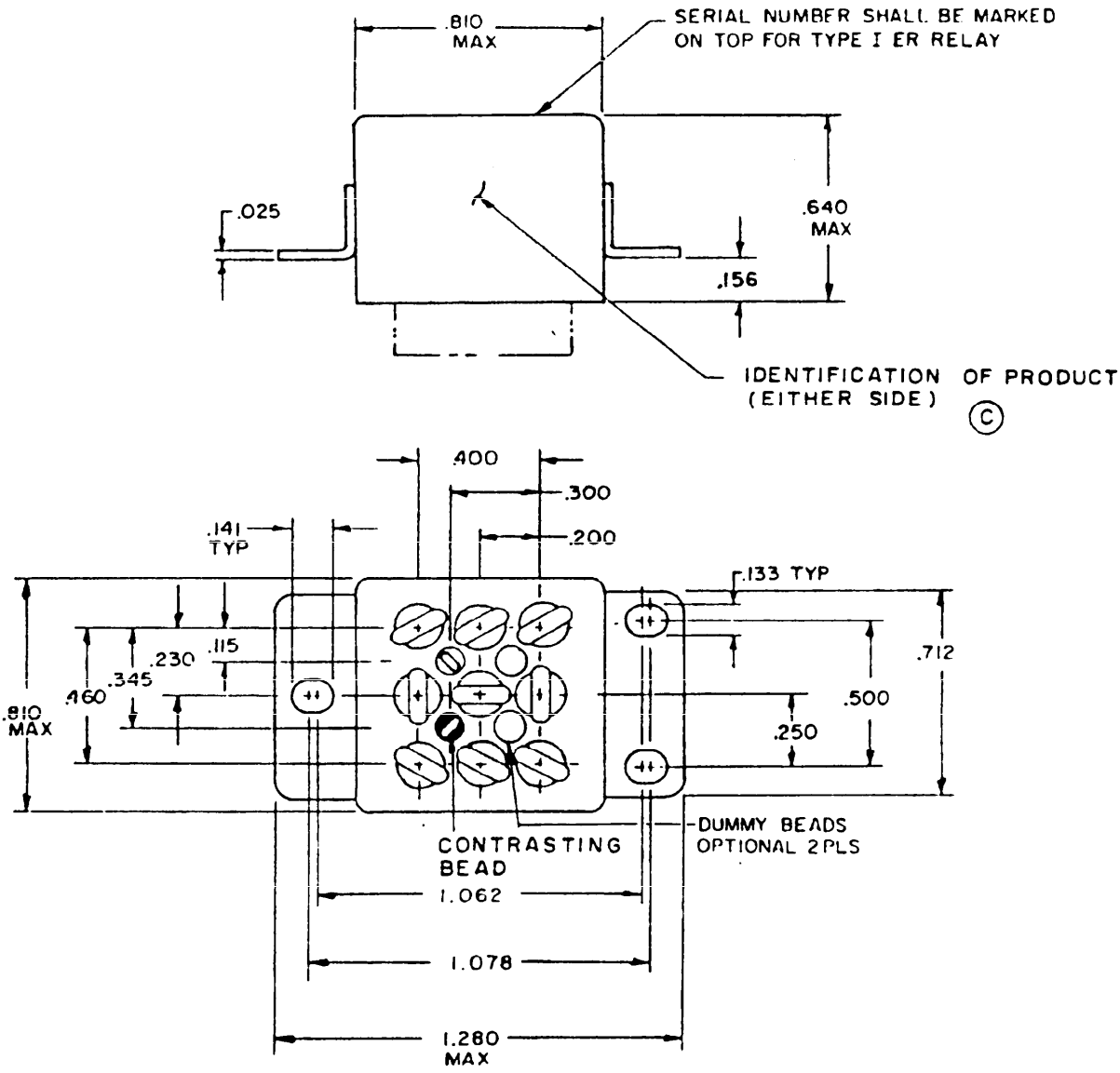
CONFIGURATION A

FIGURE 1. Outline drawing (for details, see table I).

© denotes changes

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

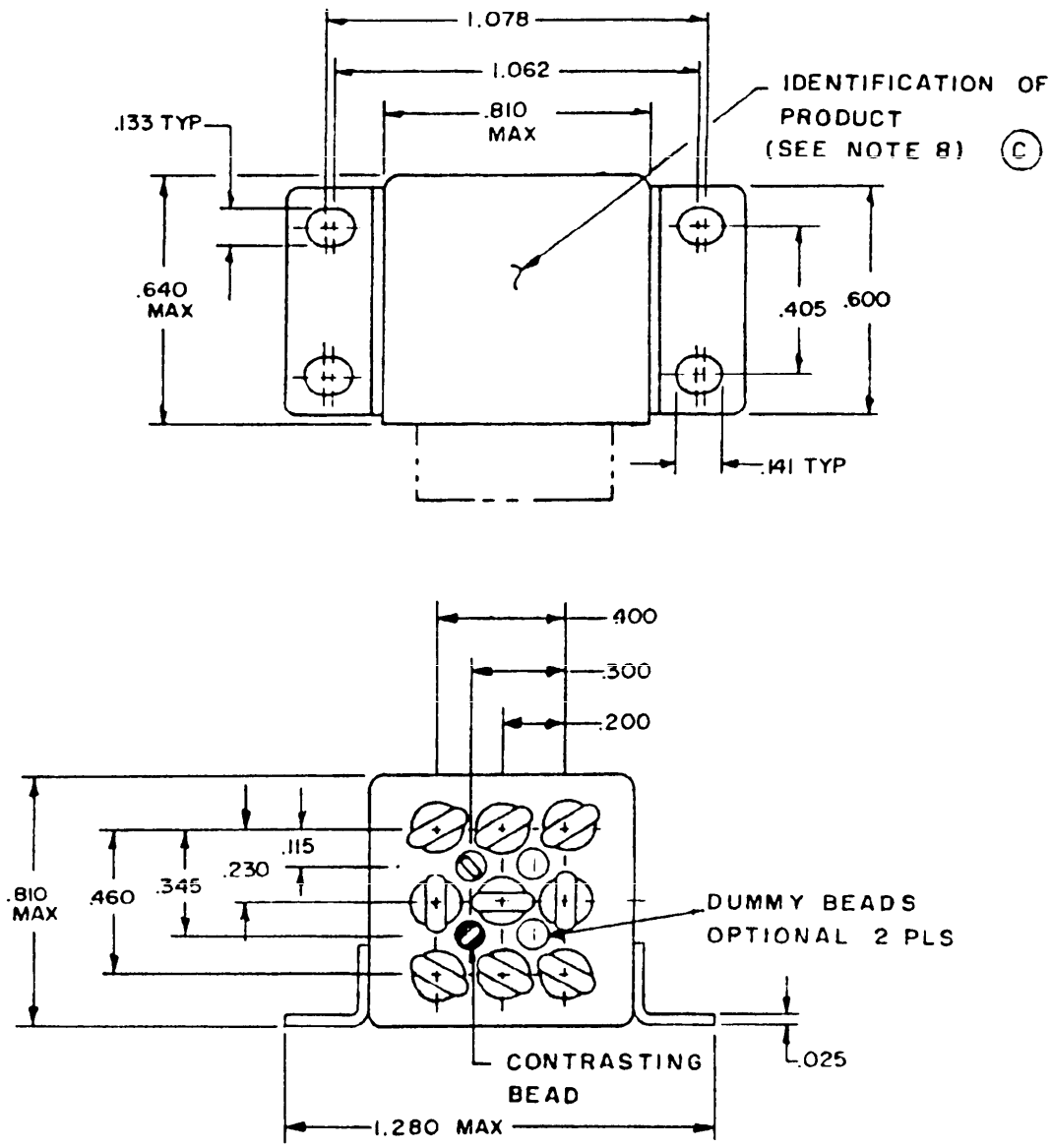
Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:28Z
Check the source to verify that this is the current version before use.



CONFIGURATION B

FIGURE 1. Outline drawing (for details, see table I) - Continued.

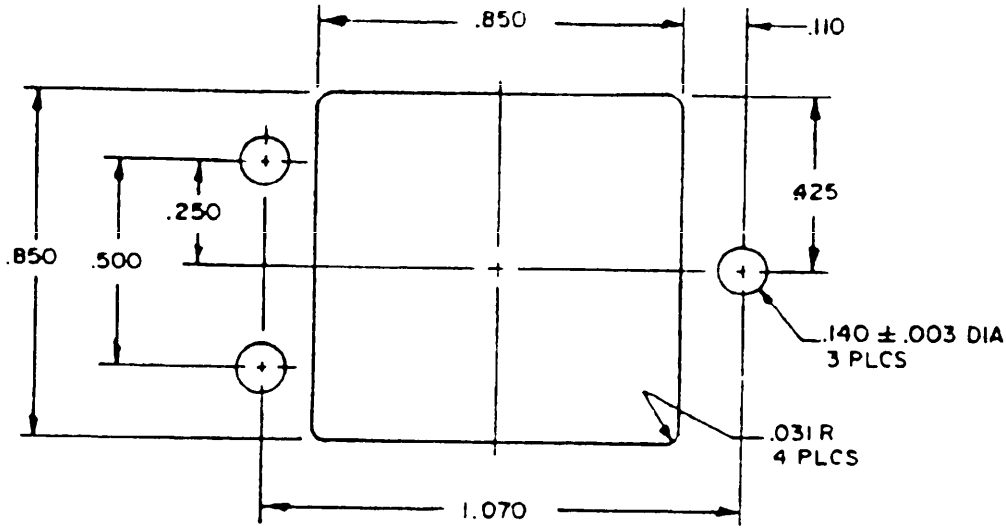
Source: https://assist.dla.mil -- Downloaded: 202610-17T06:28Z
Check the source to verify that this is the current version before use.



CONFIGURATION C

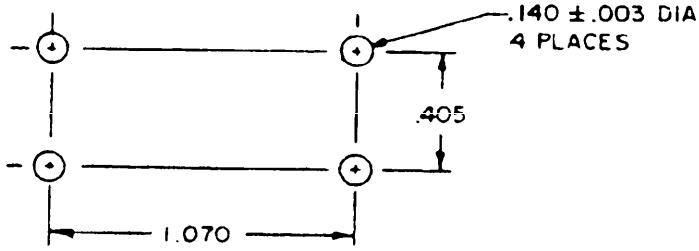
FIGURE 1. Outline drawing (for details, see table I) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:28Z
Check the source to verify that this is the current version before use.



MOUNTING HOLE LAYOUT

CONFIGURATION B



MOUNTING HOLE LAYOUT

CONFIGURATION C

FIGURE 1. Outline drawing (for details, see table I) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:28Z
Check the source to verify that this is the current version before use.

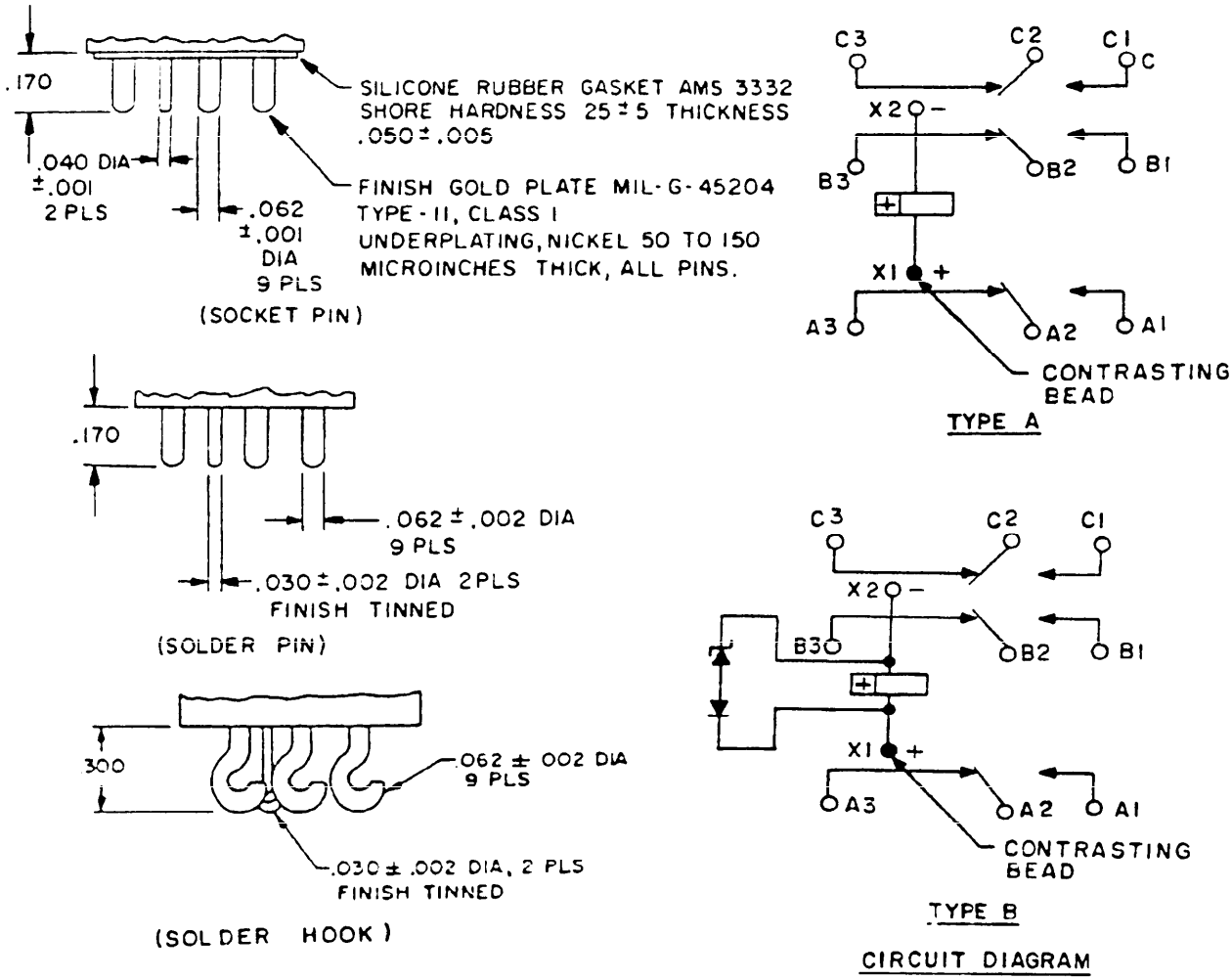


FIGURE 1. Outline drawing (for details, see table I) - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-17T06:28Z
Check the source to verify that this is the current version before use.

Inches	mm	Inches	mm	Inches	mm	Inches	mm
.001	0.03	.110	2.79	.230	5.84	.600	15.24
.002	0.05	.115	2.92	.250	6.35	.640	16.26
.003	0.08	.133	3.38	.300	7.62	.712	18.08
.025	0.64	.140	3.56	.345	8.76	.810	20.57
.030	0.76	.141	3.58	.400	10.16	.850	21.59
.031	0.79	.156	3.96	.405	10.29	1.062	26.97
.040	1.02	.170	4.32	.425	10.80	1.078	27.38
.050	1.27	.200	5.08	.500	12.70	1.280	32.51

NOTES:
1. There shall be affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified.
2. Metric equivalents are given for general information only.
3. Dimensions are in inches.
4. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
5. These relays are polarized monostable.
6. This relay shall not operate or be damaged by reverse polarity. Semiconductors shall not be used for this purpose.
7. Permanent magnet drive consists of a permanent magnet with its flux path switched and combined with the electro-magnet flux.
8. Applicable to configuration C only. The circuit diagram, manufacturer's part number, and the military part number shall be marked on the near side. The remaining portion of the nameplate data shall be marked on the far side. Relays shall be marked with the manufacturer's name or source code and data code. In addition, ER relays shall be marked with the serial number. Marking shall be with the bottom of print adjacent to the near side.

FIGURE 1. Outline drawing (for details, see table I) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:28Z
Check the source to verify that this is the current version before use.

TABLE I. Mechanical and physical characteristics. 1/

Military part number 2/	Type	Coil suppression (transient voltage, back EMF) 2/	Terminal type	Mating socket	Mounting	Circuit type
M6106/29-001	I	N/A	Solder pin	N/A	Configuration A	A
" -002	I ER	N/A	"	"	"	A
" -003	I	42 V dc max	"	"	"	B
" -004	I ER	42 V dc max	"	"	"	B
" -005	I	N/A	Solder hook	"	"	A
" -006	I ER	N/A	"	"	"	A
" -007	I	42 V dc max	"	"	"	B
" -008	I ER	42 V dc max	"	"	"	B
" -009	I	N/A	Socket pin	MIL-S-12883/ 46-01	"	A
" -010	I ER	N/A	"		"	A
" -011	I	42 V dc max	"		"	B
" -012	I ER	42 V dc max	"		"	B
" -013	I	N/A	Solder pin	N/A	Configuration B	A
" -014	I ER	N/A	"	"	"	A
" -015	I	42 V dc max	"	"	"	B
" -016	I ER	42 V dc max	"	"	"	B
" -017	I	N/A	Solder hook	"	"	A
" -018	I ER	N/A	"	"	"	A
" -019	I	42 V dc max	"	"	"	B
" -020	I ER	42 V dc max	"	"	"	B
" -021	I	N/A	Socket pin	MIL-S-12883/ 46-01	"	A
" -022	I ER	N/A	"		"	A
" -023	I	42 V dc max	"		"	B
" -024	I ER	42 V dc max	"		"	B
" -025	I	N/A	Solder pin	N/A	Configuration C	A
" -026	I ER	N/A	"	"	"	A
" -027	I	42 V dc max	"	"	"	B
" -028	I ER	42 V dc max	"	"	"	B
" -029	I	N/A	Solder hook	"	"	A
" -030	I ER	N/A	"	"	"	A
" -031	I	42 V dc max	"	"	"	B
" -032	I ER	42 V dc max	"	"	"	B

1/ For type I ER relays, a suffix letter, M, U, X, or P to indicate the applicable failure rate level shall be added to the applicable listed dash number. Failure rate level (percent per 10,000 operations), M = 1%, U = 0.5%, X = 0.3% and P = 0.1%. Example: 001M ----, 003P.

2/ JANTX or equivalent screened semiconductors shall be used in ER relays with internal coil suppression.
Relays using suppression devices shall continue to operate should the suppression circuit be in a failure mode.
Diodes shall have a peak inverse voltage of 600 V dc minimum.

REQUIREMENTS:

Coil data:

Coil data: See tables I and III.

Duty rating: Continuous.

Operational data:

Rated contact load (endurance): See table IV.

Source: https://assist.dla.mil -- Downloaded: 202610-17T06:28Z
Check the source to verify that this is the current version before use.

ive applications shall be 20 minutes

Operate time: See table III.

Release time: See table III.

Break bounce: See table III.

(C) Time current relay characteristics: Applicable (see table II). 1/

(C) TABLE II. Time-current relay characteristics at +25°C. 1/

1	20A	1 hour
2	30A	20 seconds
3	75A	2 seconds
4	150A	.53 second

1/ CAUTION: Compare with time-current characteristics of the associated circuit protective device.

Physical:

Dimensions and configuration: See figure 1.

Weight (maximum): .062 pound, 28.15 grams.

Terminal type: See figure 1 and table I.

ENVIRONMENTAL CHARACTERISTICS:	Configurations A and B	Configuration C
Temperature rating	-70°C to +125°C	-70°C to +125°C
Maximum altitude rating	300,000 feet	300,000 feet
Shock g-level	200 g	100 g
Duration	6 ms	6 ms
Maximum duration contact opening	10 µs	10 µs
Vibration - sinusoidal:		
G-level	30 g	20 g
Frequency range	70 - 3,000 Hz	57 - 3,000 Hz
Vibration - random:		
Applicable standard	MIL-STD-202	MIL-STD-202
Method 214	0.4 g^2/Hz	0.2 g^2/Hz
Test condition	IG	IE
Duration	15 min ea plane	15 min ea plane
Acceleration	15 g	15 g

ELECTRICAL CHARACTERISTICS:

Insulation resistance, initial: 100 megohms.

After life or environmental tests: 50 megohms.

1/ All relays shall withstand overload and fault currents. Relays must be able to sustain five applications (make and carry only) of power concurrently on adjacent poles at each of four different current levels in the sequence as listed in table II. Separate relays shall be tested at 28 V dc and 115/200 V ac 400 Hz 3 phase. Cooling time between successive applications shall be 30 minutes. Tests shall be performed on both normally open and normally closed contacts. There shall be no failures or evidence of welding or sticking and relays shall pass contact voltage drop at the conclusion.

Source: https://assist.dla.mil -- Downloaded: 20261O-17T06:28Z
Check the source to verify that this is the current version before use.

© TABLE III. Operating characteristics.

Dash number	Coil	Coil data: Nominal: Volts 3/	Nominal: Freq. Hz	Nominal: Res. Ω±10% @25°C	Max: Volts	Max: Amp	Max pick-up voltage 1/: Normal	Max pick-up voltage 1/: High temp test	Max pick-up voltage 1/: Cont current test	Hold voltage 1/	Drop-out voltage 1/	Time milliseconds (maximum): Operate 2/	Release 4/	Contact bounce: Main NO	Main NC	Aux NO	Aux NC	Break bounce NO only
-001 thru -032	X1 X2	28	DC	400	29	.081	18	19.8	22.5	7.0	1.5	6	6	1	1			.1

1/ Over temperature range.
2/ With nominal coil voltage.
3/ CAUTION: The use of any coil voltage less than the nominal voltage will compromise the operation of the relay.
4/ From nominal coil voltage.

© TABLE IV. Rated contact load (amperes per pole) case grounded. 1/

Type of load	Life operating cycles x 10^3	28 V dc: Main NO	Main NC	Aux NO	Aux NC	115 V ac, 1 phase: Main 400 Hz	Main 60 Hz	Aux 400 Hz	Aux 60 Hz	115/200 V ac, 3 phase: Main 400 Hz	Main 60 Hz	Aux 400 Hz	Aux 60 Hz	See appropriate footnotes
Resistive	50	10	10			10				10				
Inductive	10	6	6											
Inductive	20					8				8				
Motor	50	4	4			4				4				
Lamp	50	2	2			2				2				
Transfer load														2/
Mechanical life reduced current	200	1.25	.125			1.25				1.25				
Intermediate current	Applicable per specification													
Low level	100	Applicable per specification												3/ 4/ 5/
Mixed loads	As applicable													6/

1/ Each relay possesses high level and low level capabilities. However, relays previously tested or used (make or break) above 10 mA resistive at 6 V dc maximum or peak ac open circuits are not recommended for subsequent use in low level applications.
2/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplies at rating indicated.
3/ For low level the following shall apply:
   a. During endurance and operational reliability testing, contact load shall be 10 to 50 microamperes; 10 to 50 millivolt open circuit voltage, 100 ohm maximum contact resistance.
   b. Static contact resistance shall be performed at 50 mA maximum, 50 mV maximum.
4/ For ER relays, only the alternate low level test of operational reliability shall be used.
5/ For group A contact voltage drop test, high level testing shall be performed first, followed by low level testing. The contacts shall not make or break the high level load.
6/ Relay shall be capable of switching low level while switching any of the other rated loads on adjacent contacts.

Source: https://assist.dla.mil -- Downloaded: 2026/0-17T06:28Z
Check the source to verify that this is the current version before use.

Dielectric withstanding voltage (sea level):

	Initial	After life tests
Coil to case	1,000 V rms	750 V rms
Aux contacts	---	---
All other points	1,250 V rms	1,000 V rms

Dielectric withstanding voltage (altitude): 2/

	80,000 feet	300,000 feet
Coil to case	250 V rms	500 V rms
Aux contacts	---	---
All other points	250 V rms	500 V rms

Contact voltage drop or resistance:
High level:
Initial: 0.150 volt maximum.
After life: 0.175 volt maximum.

Low level:
Initial: 0.050 ohm maximum.
After life: 0.150 ohm maximum.

Overload current: 30 amperes dc, 60 amperes ac.

Rupture current: 40 amperes dc, 80 amperes ac.

Part number: M6106/29- (dash number from table I and a suffix letter (if applicable) indicating the failure rate level).

(C) Supersession data: See table V.

TABLE V. Supersession data.

Superseded part number M6106/29-	New part number M6106/29-
-033	-025
-034	-026
-035	-027
-036	-028

(C) Cross-reference for government logistical support: See table VI.

(C) TABLE VI. Cross-reference for government logistical support.

Part number M6106/29-	Support with part number M6106/29- 1/	Part number M6106/29-	Support with part number M6106/29- 1/
-001	-002	-017	-018
-003	-004	-019	-020
-005	-006	-021	-022
-007	-008	-023	-024
-009	-010	-025	-026
-011	-012	-027	-028
-013	-014	-029	-030
-015	-016	-031	-032

1/ The suffix letter M, U, X, or P to designate the applicable failure rate level shall be added to the applicable listed dash number. Failure rate level (percent per 10,000 cycles) M, 1.0; U, 0.5; X, 0.3; P, 0.1 (Example: -5M - - - - - - -51 P).

2/ Dielectric rating may be improved by suitable insulation of terminals and wiring after installation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:28Z
Check the source to verify that this is the current version before use.

Ⓒ Qualification by similarity: See table VII.

TABLE VII. Qualification by similarity. 1/

Military part number M6106/29-	Loads		Dynamics			Environmental
	Basic qual	ER	Sockets/Pins			
	A	D	A	B	C	A
	DC 2/ 3/	DC 2/ 3/	Configuration A 2/ 3/	Configuration B 2/ 3/	Configuration C 2/ 3/	DC 2/ 4/
001	4	3	2			4
002	4	3	2			4
003	1,2,3	1,2	1			1,2,3
004	2,3	1,2	1			2,3
005	4	3	2			4
006	4	3	2			4
007	1,2,3	1,2	1			1,2,3
008	2,3	1,2	1			2,3
009	4	3	2			4
010	4	3	2			4
011	1,2,3	1,2	1			1,2,3
012	2,3	1,2	1			2,3
013	4	3		2		4
014	4	3		2		4
015	1,2,3	1,2		1		1,2,3
016	2,3	1,2		1		2,3
017	4	3		2		4
018	4	3		2		4
019	1,2,3	1,2		1		1,2,3
020	2,3	1,2		1		2,3
021	4	3		2		4
022	4	3		2		4
023	1,2,3	1,2		1		1,2,3
024	2,3	1,2		1		2,3
025	4	3			2	4
026	4	3			2	4
027	1,2,3	1,2			1	1,2,3
028	2,3	1,2			1	2,3
029	4	3			2	4
030	4	3			2	4
031	1,2,3	1,2			1	1,2,3
032	2,3	1,2			1	2,3

1/ Reference amendment 1, MIL-R-6106, for guidelines established for determining how relays are to be grouped and ranked within each subgroup.

2/ DC suppressed coils - discretes qualify only discretes; hybrids qualify only hybrids.

3/ Terminals - socket pin type terminals are considered worst case. Solder pins and hooks may be qualified by similarity.

4/ Socket pins (with gasket seals) are considered nonsimilar to solder pins and hooks and must be tested separately.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:28Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Air Force - 85
Navy - EC

Review activities:
Air Force - 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5945-0812)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T06:28Z
Check the source to verify that this is the current version before use.